

TITLE: Indications for Peptic Ulcer Prophylaxis in the ICU

PROTOCOL: Peptic Ulcer Disease Prophylaxis in the ICU

PURPOSE: There is a variability of opinion in the appropriate management and use of SUP in the ICU. There is documented greater risk of Clostridium difficile, gastritis, pneumonia, cardiovascular events, renal failure, malabsorption, vitamin deficiency (b12), electrolyte disturbance (Ca, Mg) when using SUP in the ICU. Most alarmingly is some reported increased risk of mortality when SUP is inappropriately continued when not indicated (2). Other studies have reported increased risk of delirium, dementia, GI related CA, osteoporosis, impaired medication absorption (3).

APPROVED:



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DEFINITIONS:

- H2RA: histamine 2 receptor antagonist
- PPI: proton pump inhibitor
- TBI: traumatic brain injury
- PLT: platelets
- INR: international normalized ratio
- PTT: prothrombin time
- UGIB: upper gastrointestinal (GI) bleed
- SUP: stress ulcer prophylaxis
- ICU: intensive care unit
- TBSA: total burn surface area
- GERD: gastroesophageal reflux disease

CORE PROCEDURE:

- I. Recommended duration of treatment should be approx. 8 weeks and should not be continued unless indicated during or after hospital discharge. Even more important is daily stewardship plans when patients are downgraded or discharged from the ICU to a floor level of care.

Major and minor criteria for stress ulcer prophylaxis.

- a. Major ASHP criteria (1+):
 - i. Presence of coagulopathy defined as INR >1.5, PTT > 2 times normal, or platelet count <50 000 (2)
 - ii. Occult or overt GI bleeding lasting 6 or more days
- b. Minor criteria (2+):
 - i. Septic Shock
 - ii. Vasopressor Use
 - iii. Burns > 35% TBSA (1974 recommendation)
 - iv. TBI: GCS </10 (1989 recommendation)
 - v. Glucocorticoid therapy $\geq$ equivalent Hydrocortisone 250 mg/day (Grade D recommendations)

*There are no strong recommendations for all patients in the ICU regarding SUP for every patient. Based on the most recent consensus guidelines in critical care medicine, there are only conditional recommendations (clinician and patient dependent guidelines with moderate to low certainty of evidence) and good practice guidelines (clinician consensus but an ungraded recommendation).

Indications for SUP in the ICU should be assessed daily, and timely discontinuation should be considered paramount when assessing every ICU patient.

Indications to consider discontinued SUP:

- c. Patient no longer meets criteria
- d. Patient is on goal nutrition
- e. Patient is being downgraded from ICU level of care to IMC or Floor level of care
- f. Patient is being discharged, and the SUP was not an appropriate home medication or meets any hospital reason to continue use

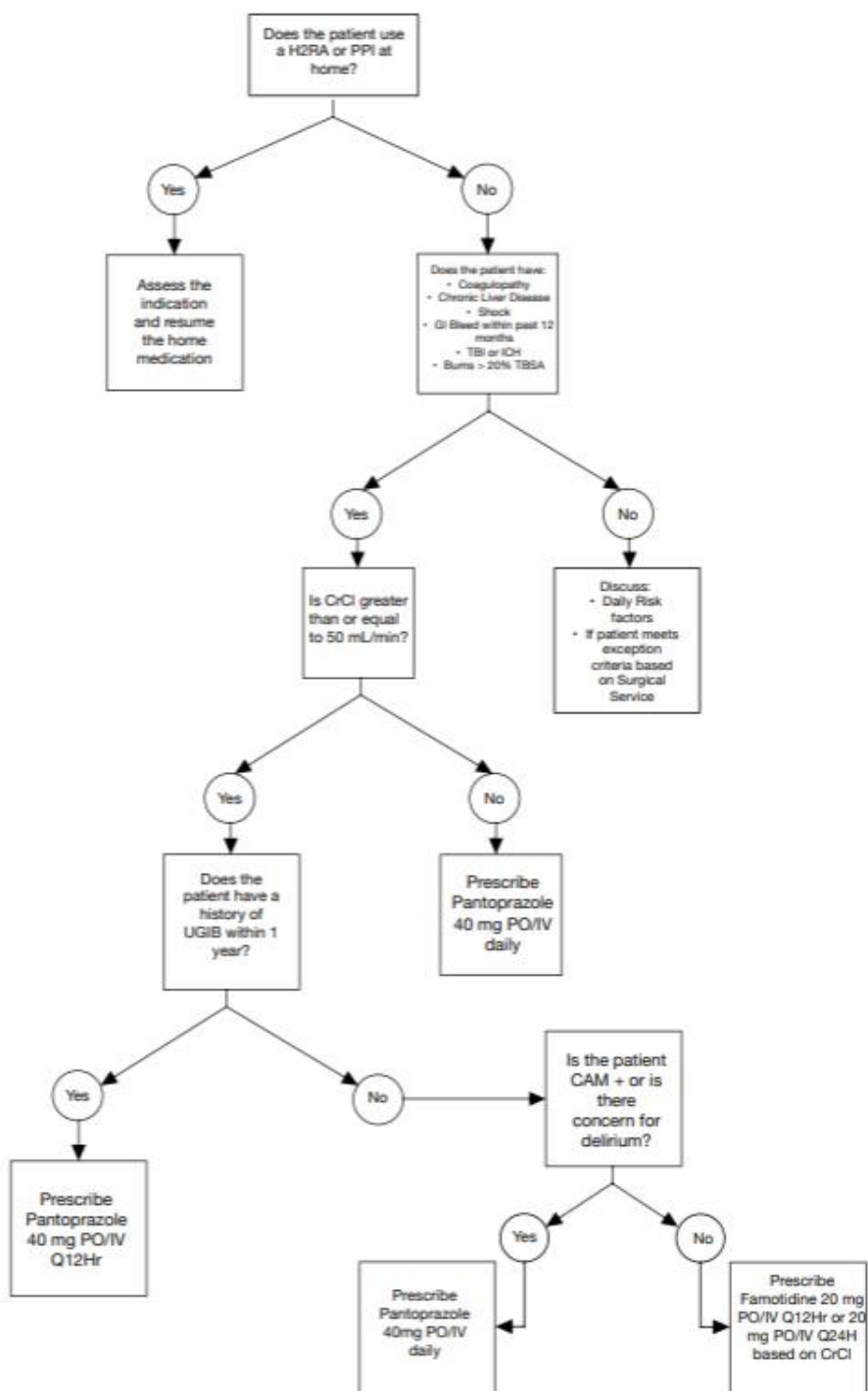
II. Long Term Side Effects of PPI Use:

- a. Pneumonia
- b. Clostridium difficile infection (C. Diff)
- c. Osteoporosis in PPI therapy
- d. Chronic Kidney Disease in PPI therapy

Suggested Flow of Decision Making when Discussing SUP in the ICU

Major Risk Factors (Must have 1 or more)	Minor Risk Factors (Must have 2 or more)
- Coagulopathy (PLT <50K, INR >1.5, PTT >2x) - History of GI bleed	- Septic shock - Vasopressor use - Multiple trauma - TBI or spinal cord injury - Burn injury >35% BSA - High dose corticosteroids: - Equivalent to Hydrocortisone $\geq$ 250 mg/day

Primary Service Requests of SUP Use	
Abdominal Transplant Surgery	PPI is preferred as patients have steroid treatment after transplant and are high risk of UGIB as they have a baseline coagulopathy.
Pancreaticobiliary Surgery	Patients will typically receive lifelong therapy of H2RA or PPI in Whipple or bypass procedures.
Ear, Nose, Throat Surgery	Patients will receive 30 days of either PPI or H2RA after total laryngectomy.
Burn Surgery	SUP in TBSA $\geq$ 20%, history of GERD, or intubated on admission.
Vascular Surgery	Patients to continue SUP post operatively and be discontinued at discharge from ICU or to home as clinically appropriate.
Minimally-Invasive Surgery	All patients after Gastric Bypass should be on life-long PPI
Thoracic Surgery	Patients after Esophagectomy should be on PPI – defer to Dr. Oduntan
Pulmonary Transplant	PPIs are utilized for long-term CLAD prophylaxis



ASSOCIATED POLICIES:

Venegas, Brittany. (2019). Evidence Based Utilization of Stress Ulcer Prophylaxis. University of Florida Critical Care Medicine Protocol. Published December 29, 2019.

KEYWORDS:

Intensive care, trauma, surgical critical care, enteral nutrition, gastrointestinal bleeding, histamine 2 blockers, gastritis, proton pump inhibitors, stress ulcer prophylaxis, critical illness, pneumonia, clostridium difficile.

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